

- $\hat{p}_n(v_i|\mathbf{x}) \neq 0$. This assumption is plausible because the output probability of an LM is usually calculated by a logistic function whose range is $(0, 1)$.
- Non-negative quantities $\alpha_{\mathbf{x}}$ and $\alpha_{\mathbf{x};v_i}$ (specific to $\mathbf{x}$) represent the discrepancies between $\hat{p}_1(v_i|\mathbf{x})$ and the true probability $p(v_i|\mathbf{x}) = \frac{N_{\mathbf{x};v_i}}{N_{\mathbf{x}}}$.
- When tracking the same specific $\mathbf{x}$ across generations, let non-negative quantities $\alpha[n]$ and $\alpha_i[n]$ denote the errors introduced during the training of the n -th generation LM. For clarity, we omit all subscripts $\mathbf{x}$ and simplify subscripts $\mathbf{x}; v_i$ to i . Specifically, $\alpha[1] \triangleq \alpha_{\mathbf{x}}$, $\alpha_i[1] \triangleq \alpha_{\mathbf{x};v_i}$. **Assume that:**
 - There exists an index i for which $\alpha_i[n]$ does not tend to zero as $n \rightarrow \infty$. This assumption is made based on the following intuition: with $\alpha_i[n]$ modeling the errors, if the training hyperparameters are the same for every LMs, rarely the error factors will not exist in the later LMs, making $\alpha_i[n]$ tend to zero when $n \rightarrow \infty$.

2.2 Main Theorem

Theorem 2.1 (LM collapse). *For any specific token sequence $\mathbf{x}$ and token v_i , along with corresponding $\alpha[j]$ and $\alpha_i[j]$, under the Replace paradigm,*

$$\forall \epsilon > 0 \exists n_0 \in \mathbb{N} \forall n > n_0 \left(\left| \hat{p}_n(v_i|\mathbf{x}) - \frac{\sum_{j=1}^n \alpha_i[j]}{\sum_{j=1}^n \alpha[j]} \right| < \epsilon \right).$$

Under the Accumulate-Subsample paradigm,

$$\forall \epsilon > 0 \exists n_1 \in \mathbb{N} \forall n > n_1 \left(\left| \hat{p}_n(v_i|\mathbf{x}) - \frac{k \sum_{j=1}^{n-1} \frac{1}{1+jk} \alpha_i[j]}{k \sum_{j=1}^{n-1} \frac{1}{1+jk} \alpha[j]} \right| < \epsilon \right),$$

where k , as defined in Section 2.1.1, is the rate at which generated data is added to the corpus.

Remark 2.2. This theorem indicates that when n is sufficiently large, $\hat{p}_n(v_i|\mathbf{x})$ can approach arbitrarily closely a term composed solely of the errors $\alpha[n]$ and $\alpha_i[n]$ introduced in each generation, rather than necessarily being close to the ground-truth distribution of the initial corpus. This may cause the cross-entropy between $\hat{p}_n(v_i|\mathbf{x})$ and $p(v_i|\mathbf{x})$ to increase, leading to an increase in the test loss of the LM on the initial corpus. This is consistent with the concept of model collapse.⁴

To prove the Main Theorem, we will derive the expression for $\hat{p}_n(v_i|\mathbf{x})$. Section 2.3 will present the expression for the initial value $\hat{p}_1(v_i|\mathbf{x})$, Section 2.4 will present the recurrence relation that relates $\hat{p}_n(v_i|\mathbf{x})$ to $\hat{p}_{n-1}(v_i|\mathbf{x})$, and Section 2.5 will solve the recurrence equation to obtain the expression for $\hat{p}_n(v_i|\mathbf{x})$. In Section 2.6, we will examine the expression for $\hat{p}_n(v_i|\mathbf{x})$ and prove that it satisfies the conditions stated in the Main Theorem.

2.3 Generation 1: initial output distribution

The training approaches for generation 1 under both data paradigms are identical, hence the following content in this section applies to both paradigms: Regardless of what value $\hat{p}_1(v_i|\mathbf{x})$ takes, it can be expressed through the combination of $p(v_i|\mathbf{x})$ with non-negative errors—as per the conventions in Section 2.1.2, we denote these non-negative quantities using $\alpha_{\mathbf{x}}$ and $\alpha_{\mathbf{x};v_i}$.

Proposition 2.3. $\forall \mathbf{x} \in V^* \forall v_i \in V$, $\hat{p}_1(v_i|\mathbf{x})$, $p(v_i|\mathbf{x}) = \frac{N_{\mathbf{x};v_i}}{N_{\mathbf{x}}}$, $\alpha_{\mathbf{x}}$, and $\alpha_{\mathbf{x};v_i}$ are related by

$$\hat{p}_1(v_i|\mathbf{x}) = \frac{N_{\mathbf{x};v_i} + \alpha_{\mathbf{x};v_i}}{N_{\mathbf{x}} + \alpha_{\mathbf{x}}}, \quad (1)$$

where

$$\alpha_{\mathbf{x}} \triangleq \sum_{v_i \in V} \alpha_{\mathbf{x};v_i}.$$

⁴Prior theoretical works on model collapse (Gerstgrasser et al., 2024; Kazdan et al., 2024; Dohmatob et al., 2024b) often proved that a generative model’s test loss tends to infinity as the number of generations grows. However, LMs have their special property—their test loss has an upper bound. This is because there is an upper bound on the cross-entropy between any two probability distributions on the vocabulary (here, the distribution predicted by the LM and the ground-truth distribution). Thus, the test loss of LMs does not tend to infinity as the number of generations grows. Therefore, we turn to the proof objective used here.

Proof. See Appendix B.1. □

2.4 Effects of per-generation errors on the output distribution of generation n ($n \geq 2$)

Subsequent LMs are trained on recursively generated data, with the specific formulation depending on the data paradigm employed. We analyze the two paradigms separately below.

2.4.1 Replace

Proposition 2.4. *Under the Replace paradigm, given $\hat{p}_{n-1}(v_i|\mathbf{x}) = \frac{y_i[n-1]}{y[n-1]}$ where $y[n-1]$ and $y_i[n-1]$ are positive scalars, $\hat{p}_n(v_i|\mathbf{x})$ admits a recursive formulation through non-negative quantities $\alpha[n]$ and $\alpha_i[n]$:*

$$\hat{p}_n(v_i|\mathbf{x}) = \frac{y_i[n-1] + \alpha_i[n]}{y[n-1] + \alpha[n]}, \quad (2)$$

where

$$\alpha[n] \triangleq \sum_{v_i \in V} \alpha_i[n].$$

Proof. See Appendix B.2. □

2.4.2 Accumulate-Subsample

Similarly, following the definition of training set composition for each generation within the *Accumulate-Subsample* paradigm, we formulate the following proposition:

Proposition 2.5. *Under the Accumulate-Subsample paradigm defined in Section 2.1.1 with accumulation rate k , given $\forall n \hat{p}_n(v_i|\mathbf{x}) = \frac{y_i[n]}{y[n]}$ where $y[n]$ and $y_i[n]$ are positive scalars, $\hat{p}_n(v_i|\mathbf{x})$ admits a recursive formulation*

$$\hat{p}_n(v_i|\mathbf{x}) = \frac{\frac{N_{\mathbf{x};v_i} + k \sum_{j=1}^{n-1} y_i[j]}{1 + (n-1)k} + \alpha_i[n]}{\frac{N_{\mathbf{x}} + k \sum_{j=1}^{n-1} y[j]}{1 + (n-1)k} + \alpha[n]}, \quad (3)$$

where

$$\alpha[n] \triangleq \sum_{v_i \in V} \alpha_i[n].$$

2.5 Solving for the output distribution of generation n

2.5.1 Replace

Equations (1) and (2) respectively present the initial value $\hat{p}_1(v_i|\mathbf{x})$ and the equation that relates $\hat{p}_n(v_i|\mathbf{x})$ to its preceding values. Combining both equations yields the following recurrence equation

$$\hat{p}_n(v_i|\mathbf{x}) = \begin{cases} \frac{N_{\mathbf{x};v_i} + \alpha_i[1]}{N_{\mathbf{x}} + \alpha[1]}, & n = 1, \\ \frac{y_i[n-1] + \alpha_i[n]}{y[n-1] + \alpha[n]}, & n \geq 2. \end{cases} \quad (4)$$

A solution is

$$\begin{cases} y_i[n] &= N_{\mathbf{x};v_i} + \sum_{j=1}^n \alpha_i[j] \\ y[n] &= N_{\mathbf{x}} + \sum_{j=1}^n \alpha[j] \end{cases},$$

$$\hat{p}_n(v_i|\mathbf{x}) = \frac{y_i[n]}{y[n]} = \frac{N_{\mathbf{x};v_i} + \sum_{j=1}^n \alpha_i[j]}{N_{\mathbf{x}} + \sum_{j=1}^n \alpha[j]}. \quad (5)$$

Finding this solution is quite straightforward. We provide a proof that this is a solution to the equation in Appendix B.3.

2.5.2 Accumulate-Subsample

Combining Equations (1) and (3) yields

$$\hat{p}_n(v_i|\mathbf{x}) = \begin{cases} \frac{N_{\mathbf{x};v_i} + \alpha_i[1]}{N_{\mathbf{x}} + \alpha[1]}, & n = 1, \\ \frac{\frac{N_{\mathbf{x};v_i} + k \sum_{j=1}^{n-1} y_i[j]}{1+(n-1)k} + \alpha_i[n]}{\frac{N_{\mathbf{x}} + k \sum_{j=1}^{n-1} y[j]}{1+(n-1)k} + \alpha[n]}, & n \geq 2. \end{cases} \quad (6)$$

A solution is

$$\begin{cases} y_i[n] &= N_{\mathbf{x};v_i} + \alpha_i[n] + k \sum_{j=1}^{n-1} \frac{1}{1+jk} \alpha_i[j], \\ y[n] &= N_{\mathbf{x}} + \alpha[n] + k \sum_{j=1}^{n-1} \frac{1}{1+jk} \alpha[j] \end{cases},$$

$$\hat{p}_n(v_i|\mathbf{x}) = \frac{y_i[n]}{y[n]} = \frac{N_{\mathbf{x};v_i} + \alpha_i[n] + k \sum_{j=1}^{n-1} \frac{1}{1+jk} \alpha_i[j]}{N_{\mathbf{x}} + \alpha[n] + k \sum_{j=1}^{n-1} \frac{1}{1+jk} \alpha[j]}. \quad (7)$$

This solution is not obvious. We provide a walkthrough of how to find this solution in Appendix B.4. We prove that this is a solution to the equation in Appendix B.5.

2.6 Completion of the Main Theorem proof

Putting Equations (5) and (7) here again, we can see that, under both paradigms, the errors (denoted by non-negative numbers $\alpha[n]$ and $\alpha_i[n]$) accumulate as n increases.

Specifically, under the *Replace* paradigm,

$$\hat{p}_n(v_i|\mathbf{x}) = \frac{\overbrace{N_{\mathbf{x};v_i} + \sum_{j=1}^n \alpha_i[j]}^{\text{positive series when } n \rightarrow \infty}}{\underbrace{N_{\mathbf{x}} + \sum_{j=1}^n \alpha[j]}_{\text{constants}}}.$$

Under the *Accumulate-Subsample* paradigm,

$$\hat{p}_n(v_i|\mathbf{x}) = \frac{\overbrace{N_{\mathbf{x};v_i} + \alpha_i[n] + k \sum_{j=1}^{n-1} \frac{1}{1+jk} \alpha_i[j]}^{\text{positive series when } n \rightarrow \infty}}{\underbrace{N_{\mathbf{x}} + \alpha[n] + k \sum_{j=1}^{n-1} \frac{1}{1+jk} \alpha[j]}_{\text{constants}}}.$$

As annotated above, under both paradigms, terms related to errors become positive series when $n \rightarrow \infty$, while terms related to the distribution of the initial corpus, $N_{\mathbf{x};v_i}$ and $N_{\mathbf{x}}$, are constants.

Taking the *Replace* paradigm as an example: Under the assumption that there exists an index i for which $\alpha_i[n]$ does not tend to zero as $n \rightarrow \infty$, it follows from the properties of positive series that the series $\sum_{j=1}^n \alpha_i[j]$ diverges. Consequently, for sufficiently large n , the difference between $\hat{p}_n(v_i|\mathbf{x})$ and the ratio $\frac{\sum_{j=1}^n \alpha_i[j]}{\sum_{j=1}^n \alpha[j]}$ becomes arbitrarily small. Formally, this can be expressed as:

$$\forall \epsilon > 0 \exists n_0 \in \mathbb{N} \forall n > n_0 \left(\left| \hat{p}_n(v_i|\mathbf{x}) - \frac{\sum_{j=1}^n \alpha_i[j]}{\sum_{j=1}^n \alpha[j]} \right| < \epsilon \right).$$

This proves the Main Theorem.